

Sanjana Singamsetty

Software Engineer / Full-Stack Developer / AI/ML Researcher

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PROFESSIONAL SUMMARY

Software Engineer with 1+ year of professional experience at PayPal building scalable, production-grade web applications using React, TypeScript, JavaScript, and modern frontend frameworks, with hands-on experience building AI-powered workflows and integrating AI/ML models into production systems. Published AI/ML researcher with 4 peer-reviewed publications (IEEE, PeerJ) across Machine Learning, NLP, Explainable AI, Healthcare AI, and Time Series Forecasting. Proven track record delivering full-stack solutions, REST APIs, and LLM-powered Agentic AI workflows using the Model Context Protocol (MCP). College Third Rank Holder and Gold Medalist in Artificial Intelligence & Machine Learning (CGPA 9.48/10). Seeking Software Engineering and AI/ML-focused roles.

CORE COMPETENCIES

Full-Stack Product Development , Applied AI/ML Research , Cross-Functional Collaboration , Production Incident Response , Agile Delivery , Stakeholder Management

Languages & Frontend	Python , JavaScript (ES6+) , TypeScript , C++ , React.js , Next.js , HTML5/CSS3 , Tailwind CSS , Redux
Backend & Data	Node.js , Express.js , REST API Development , BullMQ , Redis , MongoDB , Firebase
AI & Machine Learning	Agentic AI , LLM Integration , Machine Learning , NLP , Explainable AI , GenAI , Transformers , Hugging Face , TensorFlow , Scikit-learn , Pandas , NumPy , Streamlit
Cloud & DevOps	AWS Cloud Services , Git/GitHub , CI/CD , Docker , Vercel , Postman , Agile/Scrum
Core CS Fundamentals	Data Structures & Algorithms , System Design , OOP , Operating Systems , Computer Networks

WORK EXPERIENCE

Software Engineer

PayPal

Jul 2025 – Present

Bengaluru, India

- Delivered scalable, high-priority frontend applications under tight deadlines using React, TypeScript, JavaScript, HTML5, and CSS3, following Agile development practices.
- Served as on-call engineer, handling 4+ production incidents, performing root-cause analysis and debugging to maintain reliability of customer-facing systems.
- Built Agentic AI workflows and automation tooling using MCP, improving developer productivity and reducing manual engineering effort.
- Engineered reusable, accessible UI components aligned with WCAG accessibility standards and responsive design principles, improving long-term maintainability of the codebase.
- Collaborated cross-functionally with backend, QA, and product teams to integrate REST APIs, automate workflows, and ship production-ready features using Git, CI/CD pipelines, and Agile methodologies.

Freelance Frontend Developer

Doon International School

2025

Hyderabad, India

- Designed and developed the official school website end-to-end with a focus on responsiveness, accessibility, and user experience (UX).
- Built reusable frontend components and improved page load performance, optimizing the site across desktop and mobile platforms.
- Partnered directly with stakeholders to gather requirements, implement design improvements, and ship production-ready solutions on schedule.
- Managed full deployment lifecycle, deploying and maintaining the site in production with ongoing post-launch feature enhancements.

Cloud Engineering Intern

APSSDC

May 2023 – Aug 2023

India

- Deployed and managed cloud-hosted applications using AWS services, gaining hands-on experience with cloud infrastructure provisioning and CI/CD deployment workflows.

Flutter Developer Intern

Bragtech

May 2023 – Aug 2023

India

- Developed cross-platform mobile applications using Flutter and Firebase, implementing responsive, production-ready user interfaces for iOS and Android.

Android Developer Intern

Aerial Topographic Services Pvt. Ltd.

Mar 2023 – Apr 2023

India

- Built native Android applications with Firebase integration, implementing secure authentication and CRUD functionality.

SELECTED PROJECTS

SmartVisualizer – AI-Powered GitHub Repository Explorer

[GitHub](#) | [Live Demo](#)

- Built a full-stack AI tool that analyzes any public GitHub repository and enables natural-language chat with an AI that has read every file in the codebase.
- Implemented an interactive, color-coded file-tree Structure Explorer, commit-activity visualizations (Commit Analysis), Branch Overview with AI-generated branching strategy summaries, and language/tech-stack Code Insights.
- Developed an AI Chat agent with Deep Search for precise, file-level question answering, plus an AI-summarized README Viewer.
- Built with React, TypeScript, LLM/AI integration, and the GitHub API; deployed to production and accessible to real users.

React Learning – Full-Stack ML Web Application

[GitHub](#) | [Live Demo](#)

- Built a full-stack, ML-enabled web application using React, Node.js, and Express with secure user data handling and REST API integration.
- Implemented BullMQ and Redis for background job queuing and asynchronous task processing, improving application responsiveness.
- Deployed to production and continuously maintained, with an interactive, production-ready UI.

PUBLICATIONS (4 PEER-REVIEWED)

Comparative Analysis of Time Series Forecasting Models: Evaluating Hybrid Approaches and FB Prophet for Optimal Performance

2025

S. Singamsetty, M. Chimata, J. L. Manohar, M. S. Nandan, M. K. Enduri — CONIT, IEEE.

Enhancing Machine Learning-based Forecasting of Chronic Renal Disease with Explainable AI

2024

S. Singamsetty, S. Ghanta, S. Biswas, A. Pradhan — PeerJ Computer Science, 10:e2291.

Employing TF-IDF and Word2Vec Embeddings to Identify Multi-Class Toxicity Through Machine and Deep Learning Approaches

2024

S. Singamsetty, H. Somayajula, A. L. Immadisetty, K. S. Konkimalla, S. Tokala, M. K. Enduri — OCIT, IEEE.

Predicting Machine Learning for Early Diabetes Mellitus Prediction

2024

S. Singamsetty, M. S. Nandan, S. S. C. Polu, J. L. Manohar, S. Tokala, M. K. Enduri — BTS-I2C, IEEE.

ACHIEVEMENTS & HONORS

- Gold Medalist – Artificial Intelligence & Machine Learning, SRM AP University.
- College Third Rank Holder, CGPA: 9.48/10.
- Gold Medal, Research Day – "Enhancing Machine Learning-based Forecasting of Chronic Renal Disease with Explainable AI."
- First Prize, Mahindra University Research Symposium – patented innovation recognized for originality and impact.
- 3rd Place, CodeSprint – College-level coding competition.

EDUCATION

Bachelor of Technology, Computer Science Engineering

SRM AP University

2021 – 2025

CGPA: 9.48/10

College Third Rank Holder | Gold Medalist – Artificial Intelligence & Machine Learning

Class XII

Chaitanya College, State Board

2019 – 2021

954/1000

Class X

Atkinson Senior Secondary School, CBSE

2019

384/500